

บันทึกการปฏิบัติงานประจำ

SA STUDENT AFFAIRS
SCHOOL OF ENGINEERING

KMITL



โครงการฝึกงาน

ประจำปีการศึกษา 2568

รหัสนักศึกษา

ชื่อ - นามสกุล

ภาควิชา / สาขา



บันทึกการปฏิบัติงานประจำ

ระเบียบการฝึกงาน

INTERNSHIP REGULATIONS

ประจำปีการศึกษา 2568 Academic Year 2026

Student ID 66011390

NAME : Thit Sriprasonk

Department : R&D CPSBU Firmware Engineer

บันทึกการปฏิบัติงานประจำ Work record

บันทึกการปฏิบัติงานประจำ

ชื่อ-นามสกุล (นักศึกษา) นายธิติ ศรีประสงค์

(Full name) Thiti Sriparsonk

รหัสนักศึกษา 6601390

Student ID 66011390

ภาควิชา วิศวกรรมไอโอทีและสารสนเทศ หลักสูตร วิศวกรรมระบบไอโอทีและสารสนเทศ และฟิสิกส์
อุตสาหกรรม (สองปริญญา)

Department : R&D CPSBU Firmware Engineer **Program** หลักสูตรปกติ

Mobile No 0634385163

สถานประกอบการ : Delta Electronics Thailand PCL.

Agency name : Delta Electronics Thailand PCL.

ผู้ดูแล : Haribabu P.

(Caretaker) : Haribabu P.

ตำแหน่ง : Mentor

Position : Mentor

This image shows a full page of a handwriting practice worksheet. It features ten sets of horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

บันทึกการปฏิบัติงานประจำ

11 May : Studied Secure Boot and Secure Firmware Update using X-CUBE-SBS (UM2262).

12 May : -Studied Cyber Resilience Act (CRA) regulations.

- Studied the TFM application using STM32CubeL5 (UM2671).

13 May : -Installed programming tools for STM32 firmware.

- Built HSI clock configurations using Keil uVision.

14 May : -Read the STM32G4 Reference Manual (RM0440) and User manual STM32G4 Nucleo-64 boards.

- Built HSE clock configurations using Keil uVision.

15 May : -Made the progress presentation for the past week.

- Checked department devices.

[illegible]

บันทึกการปฏิบัติงานประจำ

18 May : -presentation to Mentor.

- department devices to calibration.

19 May :-Studied Timer of STM32G4 Reference

-Manual (RM0440).

- Build Timer PWM.

20 May : -Studied InputCapture of Timer.

- Build Timer InputCapture.

21 May : -Studied Counter Modes of Timer.

- Build Timer InputCapture.

- Test Counter Modes on PWM.

22 May : -make presentation and report.

- Accessibility: Good to go

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the width of the page. There is no text or other markings on the paper.

บันทึกการปฏิบัติงานประจำ

25 May : -Studied the ADC section in the STM32G4 Reference Manual (RM0440).

26 May : -Studied the ADC section in the STM32G4 Reference Manual (RM0440).

27 May : -Built an ADC differential trigger project using software and read the data via DMA.

28 May : -Built an ADC differential trigger project using software and read the data via DMA.

29 May : -make presentation and report for ADC.

[illegible]

บันทึกการปฏิบัติงานประจำ

8 June : -Studied trigger ADC. www

- Studied sampling ADC.

9 June : -Studied over sampling ADC.

- Studied offset ADC.

10 June : -Studied DMA.

- Studied DEMUX.

11 June : -Build ADC project.

12 June : -make presentation and report.

[illegible]

บันทึกการปฏิบัติงานประจำ

15 June : -Build DEMUX project.

16 June : -make presentation and report.

17 June : -presentation to mentor

18 June : -setting device in CPSBU. from calibration.

- Fixing issues in my DEMUX project.

19 June : -Fixing issues in my DEMUX project.

[illegible]

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22 June : -Studied USART asynchronous.

- Build USART asynchronous project.

23 June : -Studied USART multiprocessor communication.

- Build USART multiprocessor communication project.

24 June : -Build USART multiprocessor communication project.

25 June : -Studied SPI.

- Studied W5500.

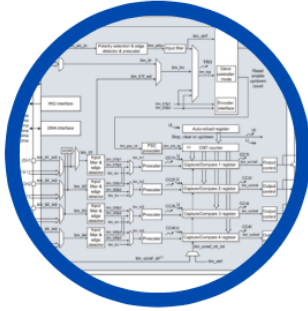
- Build SPI using W5500.

26 June : -Build SPI using W5500.

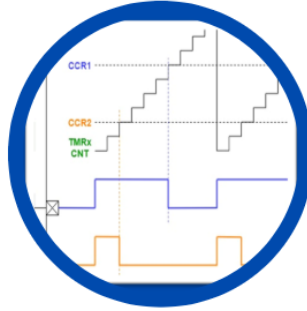
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บันทึกการปฏิบัติงานประจำ

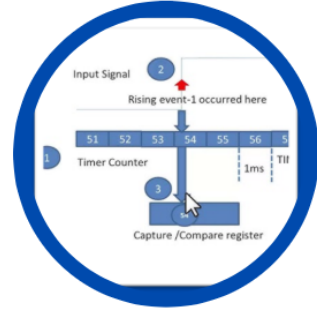
About



Timer2 diagram

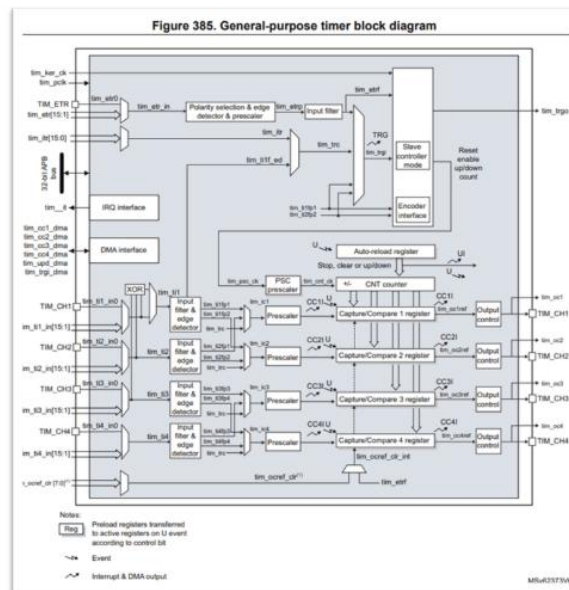


PWM Output
mode

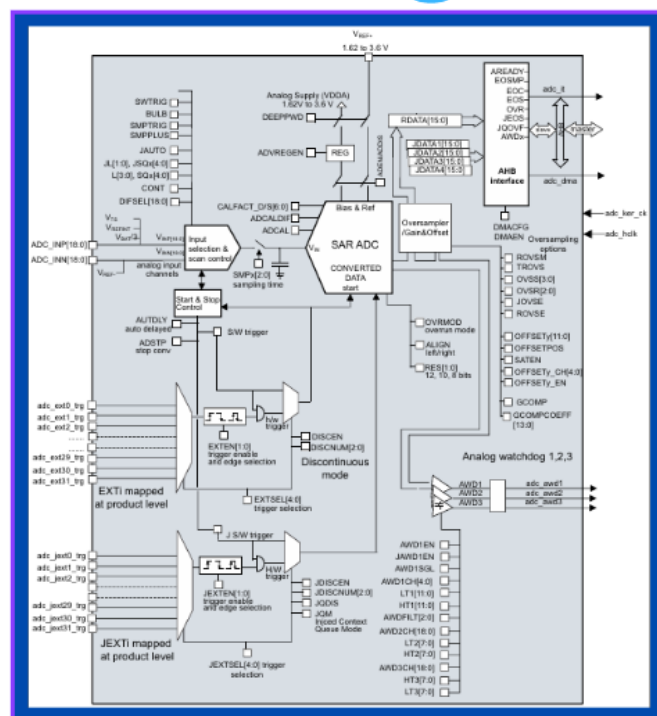


Input capture
mode

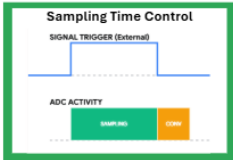
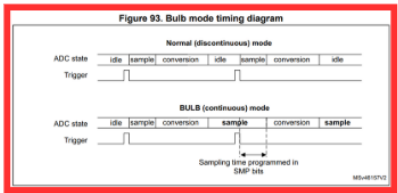
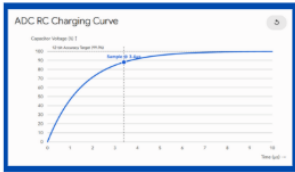
Timer diagram



PWM mode : Working



ADC Sampling Time



Channel-wise Programmable Sampling Time

- Allows custom charging time for each channel (SMPR1, SMPR2).
- Constraint: Must match the external circuit's impedance for 12-bit accuracy.

Precision Enhancements

- Voltage Booster: Internal pump to reduce analog switch resistance (essential for $V_{DDA} < 2.4V$).
- SMPPLUS Bit: Adds a fine-tuning offset (+0.5/1 cycle) for stable timing.

Sampling Control Triggers

- Bulb Sampling Mode (BULB = 1): Sampling starts immediately after the previous conversion ends. The trigger is used only to stop sampling and start the conversion.
- Sampling Time Control Trigger Mode (SMPTRIG = 1): The external trigger signal controls the exact sampling duration. A rising edge starts sampling, and a falling edge ends sampling to start the conversion.

Analog window watchdog 1

Monitoring all channels with the same limits thresholds, or focusing on just one specific channel.

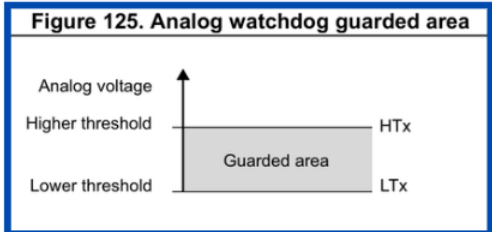
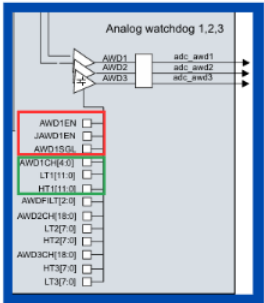
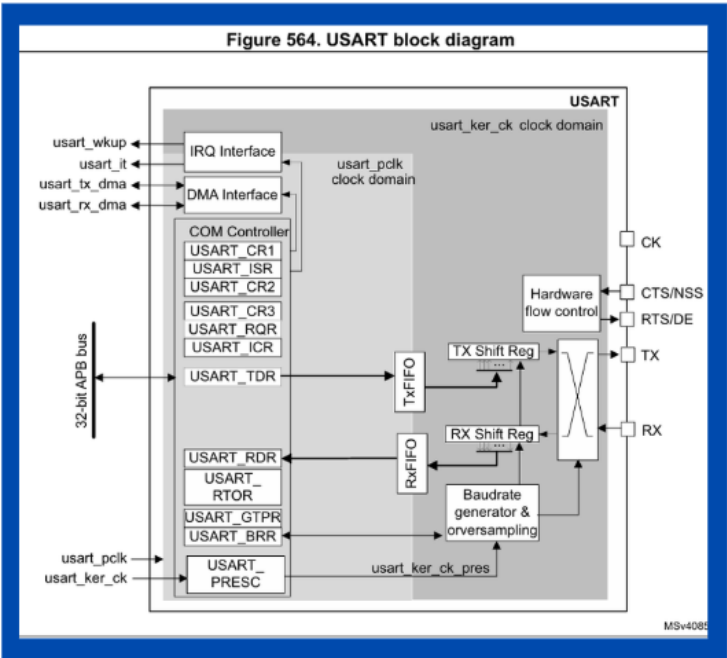


Table 172. Analog watchdog channel selection			
Channels guarded by the analog watchdog	AWD1SGL bit	AWD1EN bit	JAWD1EN bit
None	x	0	0
All injected channels	0	0	1
All regular channels	0	1	0
All regular and injected channels	0	1	1
Single ⁽¹⁾ injected channel	1	0	1
Single ⁽¹⁾ regular channel	1	1	0
Single ⁽¹⁾ regular or injected channel	1	1	1

Table 173. Analog watchdog 1 comparison			
Resolution(bit RES[1:0])	Analog watchdog comparison between:		Comments
	Raw converted data, left aligned ⁽¹⁾	Thresholds	
00: 12-bit	DATA[11:0]	LT1[11:0] and HT1[11:0]	-
01: 10-bit	DATA[11:2],00	LT1[11:0] and HT1[11:0]	User must configure LT1[1:0] and HT1[1:0] to 00
10: 8-bit	DATA[11:4],0000	LT1[11:0] and HT1[11:0]	User must configure LT1[3:0] and HT1[3:0] to 0000
11: 6-bit	DATA[11:6],000000	LT1[11:0] and HT1[11:0]	User must configure LT1[5:0] and HT1[5:0] to 000000

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USART diagram



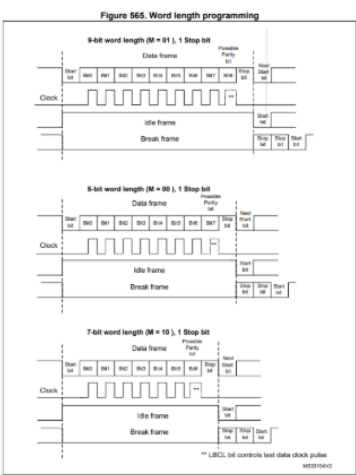
USART Frame & parity control

Table 376. USART frame formats

M bits	PCE bit	USART frame ⁽¹⁾
00	0	SB 8 bit data STB
00	1	SB 7-bit data PB STB
01	0	SB 9-bit data STB
01	1	SB 8-bit data PB STB
10	0	SB 7bit data STB
10	1	SB 6-bit data PB STB

Even Parity: The hardware calculates the parity bit so that the total count of '1's in the frame (data + parity) is an even number.

Odd Parity: The hardware calculates the parity bit so that the total count of '1's in the frame (data + parity) is an odd number.



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USART multiprocessor communication

The MSB acts as a "tag." If the tag says "Address," the Slave checks if it is being called. If it is not being called, it ignores the following data. The non-addressed devices can be placed in mute mode by means of the muting function.

